stanfordnlp / dspy Public

Code | Issues 301 | Pull requests 205 | Discussions | Actions

531 Branches 111 Tags Go to file Go to file Code

dependabot[bot] chore(deps): bump mistune from 3.0.2 to 3.2.0 in /doc...

3a7116f · 20 hours ago

.github	chore(deps): bump astral-sh/...	20 hours ago
docs	chore(deps): bump mistune fr...	20 hours ago
dspy	Update DspyGEPAResult and ...	20 hours ago
tests	Update DspyGEPAResult and ...	20 hours ago
.gitignore	feat(rlm): Add RLM Module an...	3 months ago
.pre-commit-config.yaml	fix(precommit): remove hardc...	last month
CONTRIBUTING.md	docs: add ai generated code ...	2 weeks ago
LICENSE	Add License	3 years ago
README.md	fix: broken PyPI downloads ba...	6 months ago
SECURITY.md	Create SECURITY.md (#9529)	last month
pyproject.toml	Update DspyGEPAResult and ...	20 hours ago
uv.lock	Update DspyGEPAResult and ...	20 hours ago

README | Contributing | MIT license | Security

DSPy

DSPy: *Programming*—not prompting—Foundation Models

Documentation: DSPy Docs

downloads/month 7M

DSPy is the framework for *programming—rather than prompting—language models*. It allows you to iterate fast on **building modular AI systems** and offers algorithms for **optimizing their prompts and weights**, whether you're building simple classifiers, sophisticated RAG pipelines, or Agent loops.

DSPy stands for Declarative Self-improving Python. Instead of brittle prompts, you write compositional *Python code* and use DSPy to **teach your LM to deliver high-quality outputs**. Learn more via our official documentation site or meet the community, seek help, or start contributing via this GitHub repo and our Discord server.

Documentation: dspy.ai

Please go to the DSPy Docs at dspy.ai

Installation

```
pip install dspy
```

To install the very latest from `main`:

```
pip install git+https://github.com/stanfordnlp/dspy.git
```

📜 Citation & Reading More

If you're looking to understand the framework, please go to the DSPy Docs at dspy.ai.

If you're looking to understand the underlying research, this is a set of our papers:

[Jul'25] GEPA: Reflective Prompt Evolution Can Outperform Reinforcement Learning
[Jun'24] Optimizing Instructions and Demonstrations for Multi-Stage Language Model Programs
[Oct'23] DSPy: Compiling Declarative Language Model Calls into Self-Improving Pipelines
[Jul'24] Fine-Tuning and Prompt Optimization: Two Great Steps that Work Better Together
[Jun'24] Prompts as Auto-Optimized Training Hyperparameters
[Feb'24] Assisting in Writing Wikipedia-like Articles From Scratch with Large Language Models
[Jan'24] In-Context Learning for Extreme Multi-Label Classification
[Dec'23] DSPy Assertions: Computational Constraints for Self-Refining Language Model Pipelines
[Dec'22] Demonstrate-Search-Predict: Composing Retrieval & Language Models for Knowledge-Intensive NLP

To stay up to date or learn more, follow @DSPyOSS on Twitter or the DSPy page on LinkedIn.

The **DSPy** logo is designed by **Chuyi Zhang**.

If you use DSPy or DSP in a research paper, please cite our work as follows:

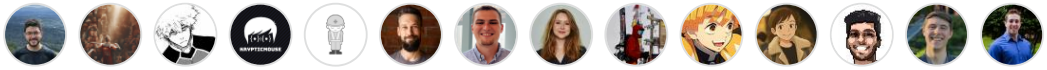
```
@inproceedings{khattab2024dspy,
  title={DSPy: Compiling Declarative Language Model Calls into Self-
Improving Pipelines},
  author={Khattab, Omar and Singhvi, Arnav and Maheshwari, Paridhi and
Zhang, Zhiyuan and Santhanam, Keshav and Vardhamanan, Sri and Haq, Saiful
and Sharma, Ashutosh and Joshi, Thomas T. and Moazam, Hanna and Miller,
Heather and Zaharia, Matei and Potts, Christopher},
  journal={The Twelfth International Conference on Learning
Representations},
  year={2024}
}
@article{khattab2022demonstrate,
```

Releases 107

3.2.0 Latest
last week

+ 106 releases

Contributors 398



+ 384 contributors

Languages

- **Python** 99.3%
- **JavaScript** 0.7%